

AYUSH SRIVASTAVA

☎ (+91)-8004662598 ✉ ayushsriva2598@gmail.com  [linkedin.com/in/ayush-srivastava2598](https://www.linkedin.com/in/ayush-srivastava2598)
 github.com/AyushSriva2598  [Leetcode - AyushSrivastava](https://leetcode.com/AyushSrivastava)

Education

KIET Group of Institution

Bachelor of Technology - Information Technology | **CGPA: 8.15**

2024 – Present

Ghaziabad, UP

Little Flower School

ISC - XII (Computer Science) | **Percentage: 93.6%**

2022 – 2023

Gorakhpur, UP

Projects

TFViz | Python, DRF, React, PostgreSQL, Docker Infrastructure as Code & GitOps Engine

2025 – Present

- Separated a Go AST parser from a Django/Neon backend into independent microservices, lowering data sync latency to under **200 milliseconds** and cutting code review time by **70%**.
- Built a **Next.js graph engine** to map complex cloud topologies in under **5 milliseconds**.
- Implemented an interactive **React Flow GitOps** pipeline rendering pre-deployment visual diffs from raw execution plans, reducing infrastructure misconfigurations and blast-radius errors by **40%** across testing suites.

API Gateway | Django, DRF, Redis, PostgreSQL, Docker, Nginx, Prometheus, Grafana, k6

2026

- Engineered a distributed API gateway in Django REST Framework + Redis 7, enforcing multi-tenant SLAs via atomic Lua scripts (Token Bucket, Sliding Window Log, Fixed Window) — sub-**1.2ms decisions**, 0 race conditions across **3 horizontally-scaled replicas** under full concurrent load.
- Load-tested to **5,000 VUs** in **k6**; fixed a **51.6%** timeout bottleneck via **Nginx keepalive pooling** and Linux socket tuning (1,024 → 65,535), sustaining sub-15ms p95 latency with exact rate-limit enforcement.
- Containerized** with **Docker** and **horizontally scaled** behind **Nginx** across stateless instances sharing **Redis** and **PostgreSQL**, with **Prometheus/Grafana** for real-time observability.

DAG Workflow Engine – Broker-less Job Queue | Django, DRF, PostgreSQL, Redis, Docker

2026

- Built a **broker-less job queue** on **PostgreSQL** using **SELECT FOR UPDATE SKIP LOCKED** with **race-safe idempotency**, letting concurrent workers claim jobs with zero double-processing or duplicate job creation.
- Implemented a **Redis token bucket** rate limiter with atomic **Lua scripting**, **exponential backoff**, a **dead-letter queue**, **graceful shutdown**, and a **stuck-job reaper** recovering orphaned jobs with zero data loss.
- Extended the queue into a **DAG-based Workflow Orchestration Engine** – reusing **SKIP LOCKED** for concurrent dispatch, with **cycle detection** and **signal-based decoupling** that auto-unlocks dependent steps on completion.

Technical Skills

- Languages:** Java, Python, JavaScript, SQL, TypeScript, HTML/CSS
- Frameworks & Libraries:** Django, Django REST Framework, React.js, Next.js, Celery, FastAPI
- Databases:** PostgreSQL, MySQL, MongoDB, Redis
- Cloud & DevOps:** AWS (S3, EC2, IAM, Lambda, DynamoDB, Glue, Athena), Docker, Terraform
- Tools & Infrastructure:** Kafka, Postman, Vercel, Linux, Git, GitHub Actions, Grafana, Prometheus, k6
- Core Competencies:** Data Structures & Algorithms, Operating Systems, Computer Networks (TCP/IP), Object-Oriented Design, Secure Coding Practices, SDLC, Agile/Scrum, REST API Design

Certifications & Achievements

- AWS Certified Cloud Practitioner**
- AWS Certified AI Practitioner**
- AWS Certified CloudOps Engineer – Associate**
- Secured **Second Position** in the **AI Prompt-A-Thon 2026**, organized by **TRYNOCODE Technology Pvt. Ltd.**

Relevant Coursework

- AWS DevOps
- System Design
- Computer Networks
- Operating Systems
- AWS CloudOps
- Database Management
- Design Analysis and Architecture